

Challenges and applications of hosting capacity analysis in DER-rich power systems[☆]

Ashish Kumar Karmaker^a, Yang Du^{a,*} , Jiajia Yang^b, Mohan Jacob^b

^a College of Science and Engineering, James Cook University, Cairns, QLD, Australia

^b College of Science and Engineering, James Cook University, Townsville, QLD, Australia

HIGHLIGHTS

- Reviews hosting capacity methods and challenges in DER-rich power systems.
- Covers both transmission and distribution level hosting capacity analysis.
- Highlights forecasting and transient constraints in hosting capacity estimation.
- Discusses applications for planning, operation, resilience, and policy design.
- Provides insights for hosting capacity-based TSO-DSO coordination.

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ABSTRACT

Increasing trends of distributed energy resources (DERs) in power system networks have led to discussions on hosting capabilities. This paper presents a structured review of academic and industrial studies from the past decade, identifying key methodological challenges and applications of hosting capacity analysis covering both transmission and distribution networks. Unlike earlier reviews that focus mainly on distribution networks, this paper addresses hosting capacity analysis for both transmission- and distribution-levels, emphasizing forecasting, steady-state and transient constraints, and application-oriented perspectives for diverse DERs. This review identifies three principal research directions: (a) role of forecasting, scenario selection, performance factors, and control strategies in hosting capacity limits, (b) consideration of steady-state and transient constraints in violation checking, and (c) translation of hosting capacity insights into practical applications for grid planning, operational scheduling, resilience enhancement, operational coordination, and market participation. By synthesizing recent academic and industrial trends, this paper offers actionable insights and strategic recommendations to empower transmission and distribution system operators and drive future applications of hosting capacity analysis in DER-centric grid infrastructures.

1. Introduction

Conventional power systems were primarily built to handle unidirectional power flows to downstream customers. The rapid proliferation of distributed energy resources (DERs), including photovoltaic (PV) systems, electric vehicles (EVs), and battery storage systems, has transformed this paradigm and intensified concerns over network hosting capacity (HC). This HC analysis offers a means to quantify the amount of additional DERs that can be accommodated without violating

operational or reliability limits, making it a necessary tool for planners and operators [1].

Hosting capacity analysis is generally applied at two levels: planning and operation. Planning HC analysis focuses on infrastructure upgrades to estimate the available capacity for future DER generation and consumption, considering long-term needs, seasonal variations, and extreme events [2]. In contrast, operational HC analysis evaluates short-term or near real-time capabilities to handle DER generation and consumption, taking into account network dynamics [3]. Although most HC studies

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* Corresponding author.

Email addresses: ashishkumar.karmaker@jcu.edu.au (A.K. Karmaker), yang.du@jcu.edu.au (Y. Du), jiajia.yang@jcu.edu.au (J. Yang), mohan.jacob@jcu.edu.au (M. Jacob).

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have focused on distribution networks, transmission-level HC is equally important, since downstream DERs can significantly affect bulk system stability and inter-regional power transfers [4].

Since the origination of the term “Hosting Capacity” in 2011 by Math Bollen, HC analysis has employed a wide range of methods, datasets, and constraints [5]. Common approaches include deterministic, stochastic, optimization-based, and streamlined methods, each balancing accuracy and computational efficiency differently [4,5]. HC analysis requires network topology, load profiles, and DER import and export profiles as inputs. The requirements and complexities of DER hosting capacity assessment depend on the chosen technologies, regulatory constraints, downstream sizes, and characteristics of the investigated networks [5]. In existing HC studies, both static and empirical time-series datasets for DERs and loads are utilized, which provide a snapshot of the current network capabilities [6,7]. These datasets fail to capture future growth, weather variations, regional customer diversities, and regulatory standards. Forecasting and state estimation techniques can address these limitations but remain underutilized in HC analysis.

Although network capacity is influenced by both steady-state and transient constraints, HC studies have primarily focused on steady-state limits such as voltage deviations, thermal loading, voltage imbalance, and power losses [8–10]. In contrast, transient constraints such as frequency stability, voltage dips and recovery, protection coordination, and dynamic voltage stability are largely unexplored in HC analysis. Research on EV hosting capacity estimation has shown that the use of multiple performance constraints may shrink hosting capacity limits and improve accuracy [11,12]. However, the computational complexity is also increased with the use of multiple constraints [11,12]. Additionally, the set of possible performance factors is theoretically open-ended (i.e., higher-order harmonics, flicker, and transient-level phenomena) [13,14]. Therefore, to achieve a balance between accuracy and computational burden, a critical selection of performance constraints through sensitivity analysis or grid-impact screening is necessary [15,16].

Table 1 presents a comparison of existing review papers published over the past decade on HC analysis, outlining DER types, performance factors, and the types of studied networks: microgrid, distribution, or transmission. It also indicates whether these studies consider application-oriented aspects, including planning and operations. The comparison shows a strong dominance of distribution-level HC studies, while transmission-level analyses are limited and mostly appear as individual primary research articles. In addition, all reviews addressed the steady-state constraints listed in Table 1. However, due to rapid fluctuations in demand and generation among participating entities, both steady-state and transient constraints are crucial for a comprehensive hosting capacity assessment, as explained in [17].

In [4,18,19], EV hosting capacity analysis, existing methods, tools, constraints, and enhancement approaches are discussed. In [5,8–10,20–24] methods, constraints, and tools available for PV hosting capacity estimation and enhancement in distribution networks are discussed. A few review papers discuss combined PV-EV hosting capacity estimation

methods [25,26], and a few explain hosting capacity enhancement in distribution networks [27]. All these papers review and discuss distribution network-centric HC analysis, overlooking transmission-level studies in DER-rich power systems.

Only a limited number of studies have investigated transmission-level hosting capacity. In [17], transmission PV hosting capacity for a 2000-bus synthetic power grid system was studied using deterministic methods that account for steady-state and transient constraints. It is shown that PV hosting capacity limits are lower when both constraints are considered to ensure system stability and reliable operation. In [28], it is demonstrated that region convexification based on the exact power flow model yields significantly higher accuracy in transmission-level hosting capacity region estimation than the DC power flow method; however, the analysis is limited to voltage stability constraints only. Additionally, a few studies have examined the transmission-distribution interface to evaluate deterministic and optimization-based hosting capacity using static and historical datasets, mostly considering voltage stability as the main criterion [11,29,30]. Hence, careful consideration of critical constraints and explicit variability of load and DERs is essential to modernize the emerging hosting capacity analysis in transmission-distribution networks.

In parallel, an industry report highlights three broader applications of hosting capacity analysis: guiding policymakers and operators, facilitating customer interconnections, and supporting strategic plans for stakeholders [31], which require a more granular application. Hence, this paper reviews HC assessment and applications in DER-rich power systems to support key stakeholders, particularly network operators and planners, in achieving mutually beneficial outcomes.

Navigating the vast body of literature on hosting capacity can be daunting for power system professionals. This review will ease the process by providing a comprehensive overview of critical challenges and applications of HC analysis in DER-rich power networks. In the current transmission networks, there has traditionally been less need for a “hosting capacity” metric in the classical distribution-level sense. However, the rapid growth of aggregated behind-the-meter DERs is now creating upstream impacts that motivate emerging work on transmission-level HC. Therefore, this review intends to consolidate hosting capacity analysis frameworks for transmission-distribution networks, with a focus on the specific contributions as follows:

- A unified review that bridges both transmission and distribution perspectives,
- Comprehensive classification of steady-state & transient constraints and the need for scenario selection and critical constraint identification,
- Role of forecasting, performance factors, prosumers, and control strategies in hosting capacity limits and
- Application-oriented HC framework linking planning, operations, resilience, market participation, and transmission-distribution coordination.

This paper addresses these gaps by presenting a structured review of academic and industrial studies from the past decade. It identifies methodological challenges, compares existing tools and approaches, and examines practical applications of HC analysis for both transmission and distribution networks. The remainder of the paper is divided into three sections. Section 2 presents background, methods, tools, and impact factors in hosting capacity analysis. Additionally, it discusses recent hosting capacity studies and challenges for both transmission and distribution networks. Section 3 summarizes the applications of hosting capacity in transmission and distribution networks. Section 4 explains future research directions, and Section 5 concludes with a summary of findings from this review.

2. Hosting capacity analysis

The term “Hosting Capacity” originated from the IT domain, specifically concerning server capacity to accommodate web user access

Table 1
Comparison with existing review papers on hosting capacity.

Ref.	PV	EV	Performance factors	Network	Applications
[5]	✓	✓	<i>S</i>	<i>D</i>	×
[4]	×	✓	<i>S</i>	<i>D</i>	×
[10]	✓	×	<i>S</i>	<i>D</i>	×
[18]	×	✓	<i>S</i>	<i>D</i>	×
[26]	✓	✓	<i>S</i>	<i>D</i>	×
[19]	×	✓	<i>S</i>	<i>M</i>	×
[9]	✓	✓	<i>S</i>	<i>D</i>	×
[22]	✓	×	<i>S</i>	<i>D</i>	×
[8]	✓	✓	<i>S</i>	<i>D</i>	×
[21]	✓	×	<i>S</i>	<i>D</i>	×
Current	✓	✓	<i>S</i> + <i>T</i>	<i>T</i> - <i>D</i>	✓

Note: ✓ - Yes, × - No, *S* - Steady-state limits, *T* - Transient limits, *M* - Microgrid, *D* - Distribution Network, *T* - Transmission Network.

[32]. Although Math Bollen introduced the term “Hosting Capacity” in 2011 to describe a network’s ability to integrate DERs [33], there are many interpretations of hosting capacity found in recent studies [4,5,10,18,26]. These definitions focus on static conditions and specific limits, instead of explicitly covering diverse technologies, dynamic behavior, and multi-index requirements. Therefore, a unified and generic definition across various technologies, methodologies, and regulatory compliance remains elusive [9,34]. Determining hosting capacity is a multidimensional problem driven by specific DERs, grid, methods, constraints, and software used. This section will review these aspects in detail to comprehend the hosting capacity analysis.

2.1. Methods

According to the literature, hosting capacity estimation can be classified based on objectives, technologies, and methods, as shown in Fig. 1. Hosting capacity analysis requires inputs with varying time resolutions, ranging from very short-term to long-term, depending on its purpose. Besides, technological differences classify hosting capacity into import, export, or both. “Import” hosting capacity is the maximum power that can be supplied to the grid, while “Export” hosting capacity is the maximum power that can be sent from the grid to distributed energy resources. Across these categories, the methods used in literature and industry projects vary, primarily falling into two groups: model-based and model-free approaches, as illustrated in Fig. 1. The model-based application requires a grid topology and associated parameters to find hosting capacity. In contrast, model-free estimation eliminates the need for an explicit grid model or extensive simulations, as applied in PV hosting capacity analysis [35,36]. Although model-free methods provide rapid data-driven responses, relying on smart devices is challenging for traditional networks, and training requires a vast amount of historical data [35]. As DERs, including PV and EV, continue to evolve and historical plug-in data remains scarce, model-free approaches still face significant challenges.

Although data-driven models are being used in the literature for HC estimation [35,36], the scarcity of DER datasets remains. The step-by-step procedure for HC estimation using a model-based approach is depicted in Fig. 2, which includes setting objectives, forecasting for obtaining load and DER datasets, DER allocation, and impact analysis. The model-based assessment methods are divided into four categories: deterministic, probabilistic, optimization, and streamlined approaches [4,5]. The deterministic method emphasizes worst-case scenarios with predefined DER data, potentially overlooking dynamic features [37,38]. The worst-case, t_{WC} , is chosen in [39], using Eq. (1), where W , $P_L(t)$ and $P_{PV}(t)$ denote the set of possible worst cases, load demand and PV export, respectively, at a given time t .

$$t_{WC} = \arg \max_{t \in W} (P_L(t) - P_{PV}(t)) \quad (1)$$

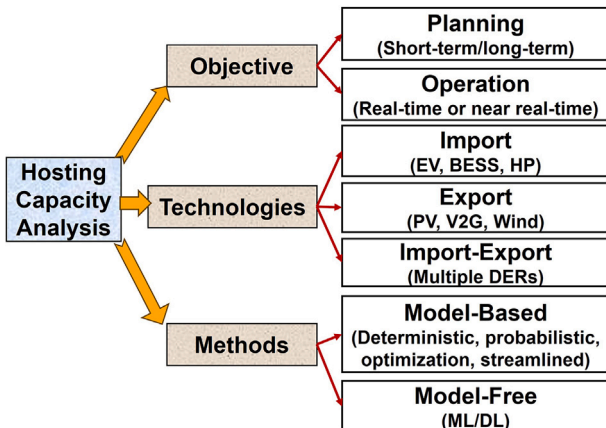


Fig. 1. Classification of hosting capacity analysis.

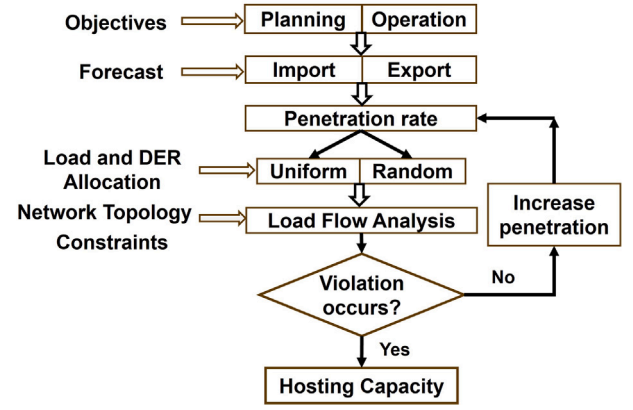


Fig. 2. Flowchart of hosting capacity analysis.

As the deterministic method reduces computational burden by skipping uncertainties with limited data, it may overlook critical events and compromise accuracy in planning and operation [10]. To capture the inherent variability of DERs, stochastic methods such as Monte Carlo simulation and Latin Hypercube sampling are widely used in scenario generation [40–42]. However, these stochastic approaches can produce inconsistent scenarios, impacting both accuracy and computational complexity [6,43]. In contrast, optimization methods aim to determine the maximum hosting capacity through optimal power flow (OPF) while adhering to grid performance for worst-cases with a specific set of constraints [44,45]. These methods are more scalable and efficient than stochastic approaches, using chance-constrained optimization to handle multiple scenarios and allow limited constraint violations within a defined probability threshold. Additionally, to address missing real-time data, forecasting and state estimation can be used to approximate unavailable inputs before executing the OPF. Moreover, selecting an inappropriate number of scenarios can distort system behavior, leading to inaccurate results, increased computational load, and suboptimal decision-making.

To address these issues, studies used a modified stochastic method called “Streamlined approach,” with a structured scenario selection that ensures computational efficiency while capturing real-world variations [12,46,47]. Firstly, PG & E formulated a streamlined approach for determining hosting capacity, facilitating better planning and operation [5]. The streamlined hosting capacity studies offer fast responses; however, scenario selection is crucial for achieving optimal accuracy. Although current studies utilize a simplistic scenario selection approach, advanced clustering or percentile-based reduction methods may be more effective in capturing the planning and operational aspects [48]. Furthermore, considering regional plug-in variations and multiple DERs is vital, yet current studies primarily focus on PV systems.

Although the core analytical steps involved in the hosting capacity analysis, such as power flow analysis, constraint checking, and scenario-based evaluation, for transmission networks are quite similar to those for distribution networks, the application granularity and operational focus differ significantly. Distribution-level HC analysis starts with downstream customers, whereas transmission-level HC starts with the aggregation of zone substations and major connection points in between T-D layers.

Distribution networks are dominated by localized steady-state constraints where affect of transient phenomena rarely limits hosting capacity. In contrast, transmission networks emphasize transient limits in addition to steady-state constraints, as disturbances at the high-voltage level directly affect system-wide stability, generator synchronism, protection limits, and inter-regional oscillations. Also, transmission networks host large synchronous generators and utility-scale inverters whose dynamic interactions require rigorous EMT or RMS stability

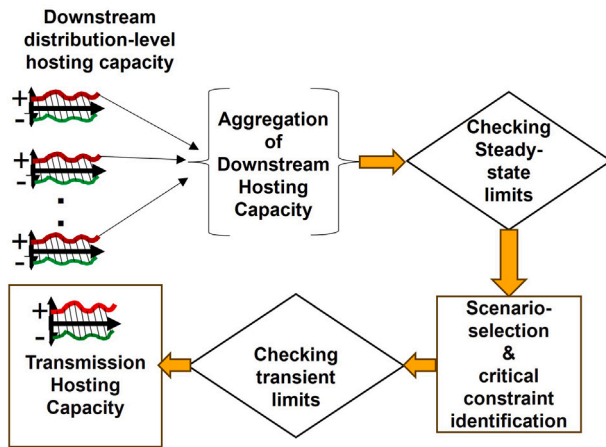


Fig. 3. Proposed method for transmission hosting capacity analysis.

assessment. Therefore, the proposed method of transmission network hosting capacity analysis may consider those mentioned factors, as shown in Fig. 3, which shows how downstream distribution-level hosting capacity profiles (import/export) are fed into the transmission-level aggregation process. First, the aggregated envelopes are checked against steady-state limits, followed by scenario selection and identifying the most critical constraint cases. These cases can be between minimum and maximum import and export profiles. The selected cases are then tested against transient stability limits, and the remaining feasible import/export envelopes can be termed as final transmission hosting capacity.

2.2. Tools

In academia and industry, several open-source and commercial tools are used for load flow analysis, as shown in Table 2, covering features such as load flow engines, stochastic models, optimization, transient analysis, customizing options, and best applications [4,21]. Typically, the selection of a tool for hosting capacity analysis depends on the objectives, data needs, technical and operational constraints, methodology, and computational time [18,25]. In the existing review papers on hosting capacity [4,5,8,10,21] tools with capabilities for distribution-centric hosting capacity were discussed. However, emerging extreme and unprecedented events on power grids reveal the importance of tools capable of steady-state capability for transient analysis.

For instance, open-source tools such as OpenDSS, GridLAB-D, and MATPOWER offer steady-state analysis with customization facilities. Among these, OpenDSS is particularly suitable for large-scale Monte Carlo simulations and hosting capacity studies, while GridLAB-D is better suited for time-series simulations and smart grid research. However, these tools generally lack transient analysis capabilities.

Besides, commercial tools such as NEPLAN, CYME, and HOMER provide a range of features, including stochastic and optimization modules. Another software, HOMER is well-suited for hybrid microgrid design and economic optimization, whereas CYME is widely used by utilities for detailed distribution system modeling. EPRI also developed “Distribution Resource Integration and Value Estimation (DRIVE)” to determine hosting capacity using steady-state voltage and thermal limits, and locational features. To integrate transient analysis, several commercial tools, such as PSCAD, PSS@E, ETAP, Powerworld simulator, and DigSILENT PowerFactory, can be used with varying degrees of customization in transmission and distribution network analysis. PSCAD is recommended for high-frequency Electromagnetic Transient (EMT) studies, while PowerFactory and PSS@E are suitable for Root Mean Square (RMS) analysis. DigSILENT PowerFactory, ideal for long-term planning, integrates most features but focuses on low-frequency transients, limiting its ability to capture high-frequency ones. PSS@E, used for both transient and steady-state analysis, employs the deterministic time-series method, which incurs a significant computational burden. Therefore, tool selection should align with the intended application, such as steady-state/dynamic studies, deterministic/stochastic approaches, and planning/operational analysis, to ensure methodological relevance and computational efficiency.

2.3. Impact factors

The impact factors are summarized in Fig. 4 and classified into grid and DER-specific factors. The consideration of more impact factors may constrain hosting capacity and increase computational burden, as explained in [4,10,26].

2.3.1. Role of grid factors

Hosting capacity is significantly influenced by grid-specific factors, such as network topology, availability of control assets, and operational constraints [5]. Grid topology affects hosting capacity by influencing power flow paths, voltage regulation, and thermal loading, especially in radial networks, which often face more constraints than meshed ones. The availability of control assets, including voltage regulators, OLTCs,

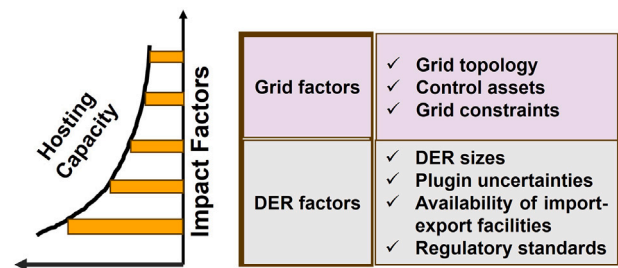


Fig. 4. Impact factors in hosting capacity assessments.

Table 2
Comparison of load flow analysis tools in hosting capacity estimation [4,5,21].

Software	Load flow	Stochastic	Optimization	Transient	Customization	Type	Best use
OpenDSS	✓	✓	×	×	Python, MATLAB	Open-Source	Planning, operation
GridLAB-D	✓	✓	✓	×	Python	Open-Source	Operational studies
DigSILENT	✓	✓	✓	✓	Python, DPL	Commercial	T-D coordination, transient
ETAP	✓	✓	✓	✓	Python, MATLAB	Commercial	Real-time operation
CYME	✓	✓	✓	×	API, Excel	Commercial	Planning with uncertainty
PSS@E	✓	✓	✓	×	Python, API	Commercial	Transmission-level analysis
NEPLAN	✓	✓	✓	×	Python, API	Commercial	Distribution planning
PowerWorld	✓	✓	✓	✓	Python, API	Commercial	Market and co-simulation
MATPOWER	✓	✓	✓	×	MATLAB, Python	Open-Source	Optimization, research
HOMER	✓	✓	✓	×	Python, Excel	Commercial	Microgrid economics
PSCAD	✓	×	×	✓	Fortran, Python, API	Commercial	High-fidelity transient

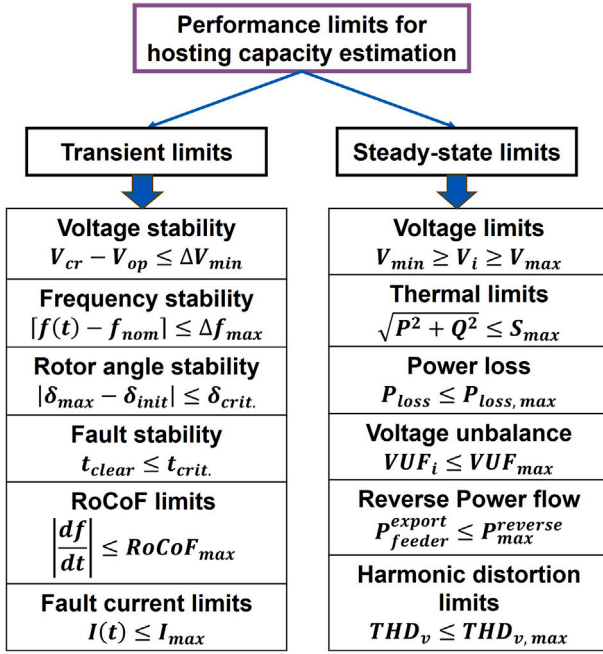


Fig. 5. Performance constraints for hosting capacity analysis.

and smart inverters, may enhance hosting capacity by actively managing voltage and reactive power, mitigating the impacts of high DER penetration.

When determining hosting capacity, grid constraints, which are divided into steady-state and transient factors, as illustrated in Fig. 5, play a crucial role. The use of single or a few selective constraints may not necessarily reflect the dominant performance indices. As discussed in [12], selecting multiple constraints improves accuracy in EV hosting capacity assessment but introduces a computational burden, requiring a trade-off between accuracy and complexity. Currently, distribution network-centric HC estimation typically considers steady-state constraints, such as voltage deviations, power loss, thermal overloading, voltage unbalance, reverse power flow, and power quality issues, with voltage limits being the most common for estimation [4,5,10]. High inverter-based renewable penetration in upstream transmission increases transient limits, including frequency deviations and voltage dips, yet these are often overlooked in planning and operation [49,50]. Unlike steady-state constraints, transient constraints evaluate system stability during disturbances and can limit hosting capacity even when steady-state limits are met [17]. Accurately capturing these effects requires EMT or RMS simulations, which are computationally intensive but essential for identifying hidden constraints in transmission-distribution networks.

2.3.2. Role of DER factors

In HC estimation, DER sizes and their allocation significantly affect localized changes in performance constraints. Homogeneous DER placement can cause local voltage or thermal issues, while random allocation may unevenly strain the grid, complicating HC assessment. Hence, improved visibility or realistic assumptions are required for accurate HC estimation, which can be achieved using high-quality datasets for generation and consumption. However, the lack of smart metering and the early stage of DER adoption make it difficult to obtain reliable datasets for accurate HC analysis.

The critical constraints and regulatory limits for hosting capacity vary across regions and network types, as shown in Table 3 [5,18]. Urban, suburban, and rural networks each have distinct topologies and operational characteristics, requiring tailored constraint selection. Urban networks should address thermal limits and voltage regulation

Table 3

Performance constraints used in different countries.

Country	PC1	PC2	PC3
Australia	− 6 % and + 10 %	100–150 %	2 %
Brazil	− 8 % and + 5 %	187.5 %	3 %
Canada	± 6 % of V _n	100 %	–
China/Taiwan	± 5 % of V _n	100 %	2 %
Denmark	± 5 % of V _n	–	2 %
United Kingdom	− 15 % and + 10 %	100 %	1.3 %
United States	± 5 %	150 %	–

PC1- Voltage limits, PC2- Overloading limits, and PC3- Voltage unbalance.

due to high customer density, while suburban networks must prioritize localized voltage deviations and thermal limits to balance current challenges and future growth. Furthermore, rural networks, with long feeders and sparse loads, should concentrate on reducing voltage drops and enhancing resilience to cope with extreme weather events. As such, explicitly incorporating the critical needs and standards allows for an accurate estimation of hosting capacity without the unnecessary inclusion of multiple constraints, which further increases complexity [62].

2.3.3. Role of forecasting

As shown in Table 4, static and empirical time-series datasets are commonly used in HC estimation, offering a snapshot of current network capabilities [58,63,64]. Due to the uncertainty and variability in load and generation, these datasets often fail to capture critical factors such as customer diversity, growth rates, weather variations, plug-in choices, and usage patterns [65]. Therefore, precise forecasting of demand and generation is needed for the grid operators to manage network performance and stability [66].

To resolve these issues, advanced forecasting models can be implemented, categorized into very short, short, medium, and long-term horizons based on the objectives for hosting capacity analysis, as shown in Fig. 6. While forecasting models are often refined to minimize errors and economic losses, their integration into hosting capacity estimation remains limited, with most models overlooking distributed demand and generation growth [67]. For instance, solar forecasting often assumes uniform irradiance across different locations, such as residential areas, suburbs, or cities. However, solar irradiance and weather patterns can vary significantly across the jurisdictional areas of large-scale power systems [68,69]. Recent literature classifies load forecasting into quantitative methods, based on empirical data, and qualitative approaches, relying on expert judgment or customer opinions [70,71]. While aggregated load is more predictable, individual demand forecasting is challenging due to variability in DER behavior, weather, and regional demographics [70]. Despite growing recognition of the importance, forecasting is not yet explicitly integrated into most hosting capacity assessment frameworks, which can lead to biased HC estimates and unnecessary computational complexity [13,72].

In the practical hosting capacity assessment framework mentioned in Fig. 2, forecasting is used as a pre-processing step to generate load and generation profiles. Specifically, forecasted demand and generation profiles are used to construct representative scenarios for deterministic HC, reduce the dimensionality and sampling space of probabilistic scenarios in stochastic HC methods, and define boundary conditions for optimization-based HC calculations. Compared with conventional Monte Carlo-based stochastic sampling, forecast-based scenario formulation simplifies HC analysis workflows and reduces computational complexity without sacrificing realism in system variability.

2.3.4. Role of prosumers

The increasing presence of prosumers introduces significant fluctuations in power exports and imports, driven by the variability of rooftop PV generation and the time-dependent charging behavior of EVs, respectively. In PV-rich distribution networks, high levels of solar

Table 4
Summary of hosting capacity studies.

Ref. & Year	Network	HC Method	DER & Load datasets	Type of DER	Constraints
[39], 2024	D	Stochastic	Empirical	PV	VL, TL, OL
[16], 2024	D	Data-driven	Empirical	PV	VS
[56], 2024	D	Optimization-based	Typical	PV	VL, TL
[57], 2024	D	Stochastic and optimization	Random	PV	VL, TL, OL
[12], 2023	D	Streamlined	Historical	EV	VL, TL, PL, VU
[38], 2022	D	Deterministic and stochastic	Random	PV	VL, VU, OL
[58], 2021	D	Optimization-based	Historical	PV	VL, PB
[7], 2021	D	Deterministic	Historical	PV-EV	VL
[59], 2021	D	Stochastic	Historical	PV	VL
[17], 2020	T	Time-series	Dynamic model	PV	TSL
[60], 2024	T	Time-series	Historical	PV	TLO
[28], 2024	T	DC power-flow	–	PV	VL, TL
[29], 2023	T-D	Optimization	Historical	PV	VL, TL, SCL
[11], 2021	T-D	Deterministic	Static	PV	VL
[30], 2020	T-D	Deterministic	Historical	PV	VL, TL, VSI
[61], 2015	T	Deterministic	Static	PV	HDL

Note: VL-Voltage limits, TL-Thermal limits, VU- Voltage unbalance, PL- Power loss, PB- Power balance, OL-Overloading limits, VS- Voltage sensitivity, VSI- Voltage stability index, HDL- Harmonic distortion limit, SCL- Short-circuit limit, TSL- Transient stability limits, TLO-Transmission line overloading.

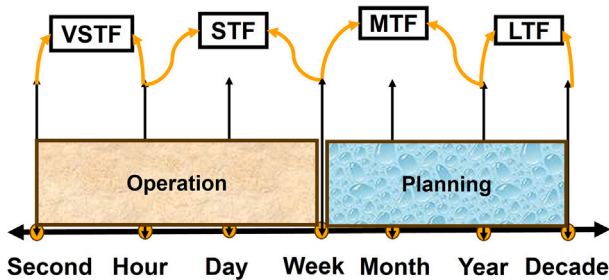


Fig. 6. Forecasting techniques for operation and planning hosting capacity estimation (VSTF- Very Short-term Forecasting, STF- Short-term Forecasting, MTF-Medium-term Forecasting, and LTF - Long-term Forecasting).

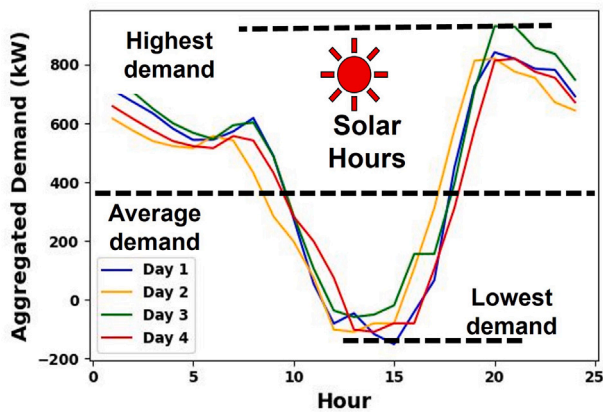


Fig. 7. Duck curve for a real PV-rich Australian distribution feeder.

PV exports, largely driven by prosumers, lead to significant generation-demand mismatches, commonly known as the “Duck Curve,” as shown in Fig. 7. It illustrates the duck curve over four consecutive summer days in an Australian distribution network, highlighting the recurring impact of high midday solar generation [73]. The recent blackout in Spain highlights the critical importance of effectively managing prosumers in hosting capacity analysis, as unregistered or poorly coordinated distributed solar generation can cause rapid fluctuations in net load and compromise grid stability through unexpected frequency deviations. However, increasing trends of V2G prosumers will pose extra stress in

the near future, shrinking export capabilities, which is evident in present studies [74,75]. Stability issues from solar and V2G exports, combined with rapid DER fluctuations from solar intermittency and EV charging, can trigger over-voltage, reverse power flow, and transient instability, reducing HC and straining protection systems [76,77]. To address these challenges, identifying potential prosumers in the network, as explored in [73,78], helps predict export probabilities and enhances the accuracy and effectiveness of HC calculations.

2.3.5. Role of control strategies

While hosting capacity is restricted by technical limits such as voltage, thermal loading, stability, and protection constraints, the extent to which these limits are reached depends strongly on how the network is operated and controlled. Therefore, HC is not a static attribute of the grid, but a dynamic outcome influenced by the effectiveness of control mechanisms across different voltage levels [79]. Fig. 8 shows the different categories of control mechanisms in transmission-distribution networks for performance management. These strategies obtained from existing studies [51–55] are divided into four categories, such as voltage control, thermal & power flow control, stability control, and flexibility & operational control strategy.

In distribution networks, control strategies primarily aim to manage localized constraints arising from customer-level DER integration [51,80]. Smart inverter functions, such as Volt-VAR, Volt-Watt, constant power factor, coordinated operation of on-load tap changers and capacitor banks, and flexible demand control, all help mitigate voltage rise, reverse power flow, and thermal stress on feeders [81]. These actions can improve hosting capacity limits by expanding feasible operating regions in distribution networks. Emerging approaches, such as dynamic limits, may further enhance these control actions by adjusting exports and imports in real time based on network conditions.

In transmission networks, control strategies operate at a broader system-level, shaping HC by maintaining stability, managing thermal congestion, and preserving security margins [55,82]. Synchronous condensers, STATCOMs, SVCs, and other flexible AC technologies enhance system strength, voltage stability, and controllability of power flow. However, the cost-effectiveness and technical viability of placement need to be assessed properly. Various schemes, such as wide-area monitoring, frequency response and ancillary services, and coordinated generation control, help ensure the system remains secure under dynamic contingencies, even with higher DER penetration at connection points [52,82]. These controls directly influence transfer capabilities, fault levels, stability margins, and expand the hosting capacity of the transmission assets. The control strategies at both levels extend

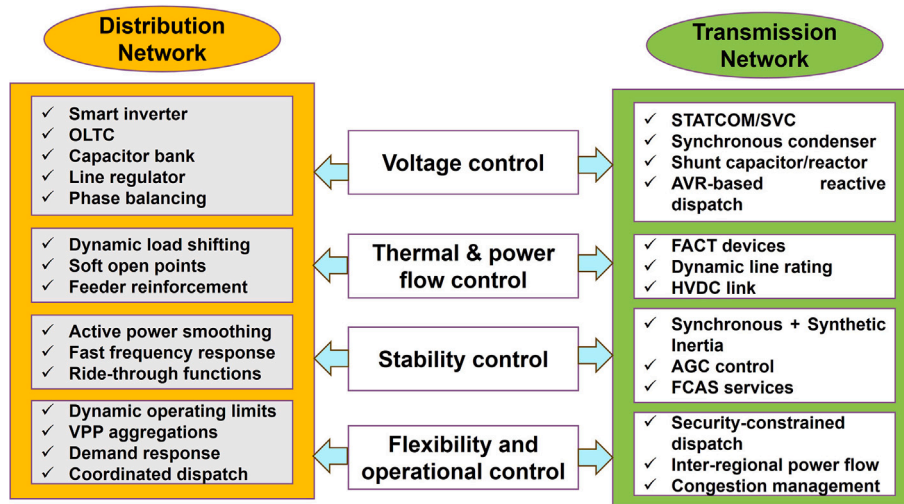


Fig. 8. Control strategies for transmission and distribution networks [51–55].

operational limits, enabling greater DER accommodation without requiring immediate network augmentation. Therefore, integrating control strategies through cost-effective feasibility studies is necessary to increase HC in T-D networks.

2.4. Challenges in existing hosting capacity studies

HC is analyzed through assessment and enhancement in recent literature, both of which rely on estimation as a foundational step. In searching Google Scholar and Scopus databases by “Hosting Capacity” for the duration between 2014 and early 2025, 1061 research papers were found for hosting capacity estimation and enhancement. These were screened to retain only peer-reviewed English-language studies that explicitly address HC estimation, enhancement, or applications in transmission and distribution, considering individual or mixed DERs.

It is seen that 72.31 % with PV, 26.29 % with EV, 0.48 % with heat pumps (HP), and 0.57 % with multiple DERs. Table 4 shows a detailed review of recent HC studies, including methods, networks, datasets, and constraints used. The majority of HC studies focus on distribution networks, with only a few studies referring to transmission-level analysis. To find the impacts of downstream DERs in upstream transmission networks, analyzing both steady-state and transient constraints is crucial, as explained in [50,83,84]. However, these studies mainly focus on stand-alone transient impacts, without integrating steady-state and transient effects to assess DER HC in transmission networks. Also, most studies focus on PV systems, with limited research addressing multiple DERs in assessing HC for transmission-distribution networks.

There are several common challenges, such as modeling complexity, variability in DER and loads, a lack of standardized methods, and the scenario selection process for HC analysis in transmission and distribution networks. Radial and unbalanced distribution networks face challenges in modeling due to limited data availability and proximity to diverse DER interconnections. Also, in transmission network HC estimation, considering inertia and system stability, including steady-state and transient limits, is crucial due to the presence of inverter-based resources. The wide-area transmission network introduces multiple constraints, including N-1 contingency handling and inter-regional flows. Transient analysis in HC studies requires complex time-domain simulations, which are computationally intensive and pose challenges for stochastic analysis, necessitating efficient scenario reduction techniques. Another challenge lies in coordinating transmission HC studies with downstream distribution networks. Moreover, market and dispatch

uncertainties may influence frequency variations and HC under complex operational and regulatory conditions.

2.5. Underlying challenges of hosting capacity analysis

Table 5 summarizes the key challenges and maturity levels for HC assessment in transmission and distribution networks. The maturity levels are semi-qualitative indicators based on a combination of factors, including methodological development, availability of tools and datasets, computational efficiency, and practical applications. Maturity values range from 0 (conceptually new) to 1 (well-established or widely adopted), indicating a comparative view of HC readiness levels across transmission and distribution networks.

Although computational challenges in probabilistic hosting capacity assessment are higher in both transmission and distribution networks, they are particularly significant in transmission networks due to several unique factors. First, transmission network studies usually integrate both steady-state and dynamic contingencies. The dynamic contingencies, such as transient stability, frequency response, protection limits, and N-1 criteria, all require time-domain simulation for each probabilistic scenario. This causes a large scenario space, which incurs computational complexity. Secondly, transmission networks are highly meshed, with inter-regional coupling that precludes local linearization or feeder-level decoupling techniques often used in distribution HC studies. Thirdly, uncertainty at the transmission level typically involves large renewable farms, bulk industrial customers, market dispatch behavior, and downstream aggregated exports/imports, which require more sophisticated stochastic modeling than distribution networks. These factors collectively place stress on HC assessment, its complexity, and accuracy in transmission networks.

Along with the computational complexity, transmission level HC studies still face challenges due to a lack of detailed downstream network models. Transmission system operators (TSOs) usually see only aggregated DER injections from downstream distribution feeders, without access to the detailed plug-in behavior for the behind-the-meter DERs. High DER penetration introduces fast inverter dynamics, voltage oscillations, and power quality transients [85]. Additionally, transmission-distribution interactions are not fully captured in standard simulation tools, such as PSS/E or PSCAD, which are primarily designed for bulk-system stability and transient studies [86]. Probabilistic HC studies are computationally intensive, and inter-regional flows and operational constraints further limit practical implementation.

Table 5

Critical challenges and maturity levels for HC analysis in transmission and distribution networks, obtained from studies over the past decade. For maturity levels, Top (green) bar: Transmission network; Bottom (blue) bar: Distribution network.

Dimension	Transmission Networks	Distribution Networks	Maturity
Modeling	- Lack of HC-specific models capturing DER impacts and T-D interactions - Accurate dynamic modeling of inverter-based resources for HC analysis	- HC requires detailed feeder/LV representation - High diversity of feeder topologies and DER types complicates modeling	
Methodology	- HC studies mostly rely on worst-case or contingency approaches - Probabilistic HC at transmission level is conceptually new and computationally heavy	- Deterministic HC well-established but conservative - Stochastic HC is more advanced, but scenario selection is complex	
Data and Tools	- PMU/SCADA data resolution limits dynamic HC studies - Low visibility for downstream DER profiles - Existing transmission tools are tailored for stability studies rather than explicit HC analysis	- Limited smart meter and DER operational data constrain realistic HC - Distribution tools (OpenDSS, CYME, GridLAB-D) include HC features but lack scalability and interoperability with transmission-level platforms	
Computational Complexity	- Probabilistic HC under stability constraints is computationally intensive - Full-scale transmission HC assessment remains impractical due to nonlinear optimization and scenario explosion	- HC estimation with high DER penetration is data- and computation-heavy - Real-time HC assessment not yet feasible for operating envelopes due to high DER diversity - Scaling HC calculations across multiple feeders with diverse DER types remains challenging	
Regulatory and Policy Constraints	- No explicit HC standards or interconnection codes at transmission level - Inter-regional transfer limits may restrict HC allocation	- HC embedded in local interconnection/export rules, but varies widely - Policies, tariffs, or restrictive limits can unintentionally constrain hosting potential of DERs	
Power Quality / Reliability	- HC impact on oscillatory stability and bulk reactive power support is not well-studied - DER fault contributions challenge transmission protection schemes	- HC constrained by voltage rise, harmonics, and imbalance issues - Reliability indices (SAIFI, SAIDI) increasingly influenced by DER	
Geospatial / Topology Factors	- Long-distance inter-regional flows affect HC distribution - Methods for allocating HC spatially at the transmission scale are limited	- Feeder type (urban/rural/suburban) and characteristics strongly influence HC - Uncertain DER locations and sizes cause voltage and thermal constraints to be more difficult to predict in HC analysis	
Economic / Market Considerations	- No clear framework for linking HC with wholesale energy market participation - DER-driven flexibility services not yet reflected in HC metrics	- HC already shaped by tariffs, incentives, and DER market pilots - Economic constraints can limit practical HC despite technical headroom	
Uncertainty / Forecasting	- Transmission HC under uncertainty is largely unexplored - Forecast-driven HC methods emerging but not validated in the transmission scale	- HC estimation under DER uncertainty is relatively complex - Requires scenario generation and validation	
Scalability and Integration	- HC concepts not integrated into transmission planning and operational workflows - Methods must extend across T-D boundaries for system-wide HC	- HC integration with distribution network management is ongoing, but still limited to pilots - Scaling across multiple feeders remains challenging	

At the distribution level, HC estimation is more advanced but still faces significant hurdles. Real-time HC assessment is theoretically possible, but the combination of high DER diversity, uncertain locations, detailed time-series data requirements, and plug-in uncertainties

makes it impractical. Distribution-level tools, such as OpenDSS, CYME, and GridLAB-D, can model feeder-level HC, but scaling across multiple feeders with upstream networks and integrating with real-time operational platforms remains challenging.

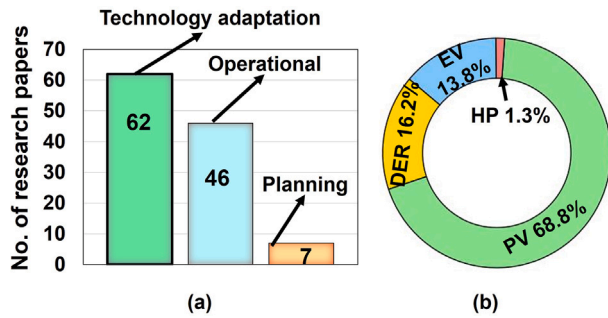


Fig. 9. Hosting capacity enhancement studies: a) types, b) technologies.

3. Applications of hosting capacity analysis

This section examines practical applications at a granular level, based on DER hosting capacity estimation, primarily categorized for planning and operational purposes.

3.1. Increased DER integration capabilities

This section reviews the key applications of HC analysis to foster future DER integration. Current research statistics identify three key approaches to enhance hosting capacity: planning, operational, and technology adaptation methods, as shown in Fig. 9(a). The effectiveness of these methods depends on stakeholder benefits, technology readiness, network reliability, and regulatory compliance [5]. Fig. 9(b) highlights the greater PV hosting capacity enhancement research in the literature compared to the EVs. The growing shift toward transport electrification has driven the need for research on combined DER hosting capacity enhancement [7].

The technology adaptation deals with the reinforcement of advanced technology in the networks. These techniques integrate OLTC transformers [87], soft open points [88,89], and feeder reconfiguration [90], to comply with voltage and thermal limits. Real-time grid management and control strategies to optimize hosting capacity without major upgrades are known as operational strategies. It includes coordinated active/reactive power management [91,92] and smart inverter control [93] to improve hosting capacity. As these methods often involve costly adjustments and technology adaptations, prioritizing planning-level solutions is pivotal to minimize future retrofits and protect stakeholder benefits. In [94], optimization algorithms are used to allocate PV resources and strengthen hosting capabilities. To improve PV hosting capacity, several planning-level solutions, including optimal placement of battery storage systems [95], and static VAR compensators [96], are proposed. To improve EV hosting capacity, the optimal charger allocation [97] and capacitors [98] are proposed. All these factors are directly linked with regional demographics, policy frameworks, and initiatives [18].

3.2. Managing prosumers' flexibilities

Managing increased prosumers in the distribution networks employs two control mechanisms—fixed and dynamic limits [99], as explained in Fig. 10. Utility operators in Australia and several other countries

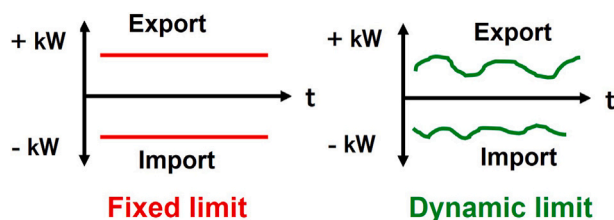


Fig. 10. Export-import control limits [74,102].

Table 6

Comparison of hosting capacity allocation methods.

Method	Complexity	Adaptability	Fairness	Forecasting
Fixed [74]	Low	No	Low	Not needed
Dynamic [102,103]	High	Yes	Medium	Load and DER
Priority-Based [73,75]	High	Yes	High	Load and DER
Market-Based [105]	High	Yes	Medium	Price

assign fixed export or import limits to customers at connection points, regardless of the diverse needs, capacities, and network constraints [74]. Setting fixed limits for solar and V2G prosumers ensures equal export opportunities but may compromise network integrity, limiting flexibility and optimal resource utilization [100]. To resolve these issues, “dynamic operating envelope (DOE)” came to light, which provides time-varying limits for the customers while maintaining technical constraints [101]. The DOEs are usually computed in terms of hosting capacity using either an iterative [101] or OPF method considering voltage and thermal limits [102]. Although it offers flexible export opportunities for prosumers, ignorance of diverse capabilities and nodal situations limits stakeholder benefits. Table 6 shows a summary of the hosting capacity allocation methods in existing literature. To maintain fairness in prosumer exports, weighted proportional [75], and equitable allocation [103], dynamic export limits and community storage systems [104] are proposed. Recently, a Shaped Operating Envelopes (SOEs) have been introduced to refine DOEs by adding aggregator/customer preferences and network support [105]. However, diverse prosumer capacities and nodal constraints may affect fairness issues. As export limits are primarily determined for PV-based battery storage, adding V2G requires further analysis, necessitating an evaluation of market trends to identify potential compatibility issues.

3.3. Resilience and reliability planning

Hosting capacity analysis can play a pivotal role in resilience and reliability planning, especially for extreme weather events and grid uncertainties. Extreme events, such as heatwaves, storms, floods, or wildfires, can significantly impact power generation and consumption patterns [106]. The hosting capacity during extreme events can be affected by temperature, wind speed, flooding, and equipment failures. The combined effect of these factors may reduce hosting capacity by a factor known as the reduction factor, R_f , as in equation (2) in Fig. 11. This reduction factor provides insights into the extent to which the grid can support customers during extreme events or vice versa. The hosting capacity under extreme events, HC_{extreme} , can be calculated as in equation (3), as shown in Fig. 11. HC_{base} is the baseline hosting capacity, which can be estimated using the methods discussed in Section 2. Each extreme event contributes a reduction factor R_i , where N is the number of extreme event factors. Fig. 11 shows the conceptual steps for improving network resilience during extreme events through hosting capacity analysis. The overall hosting capacity can be a factor of baseline hosting capacity and reduction factors based on extreme events. Network operators can enhance grid resilience by assessing hosting capacity for extreme cases to identify weak areas or optimize DER dispatch. Additionally, hosting capacity estimation helps prevent blackouts and plan for restoration by identifying DER potential in a microgrid, thereby supporting critical loads during outages [107].

3.4. Operational scheduling and market regulation

HC analysis is crucial in scheduling DER import-export to ensure grid stability and optimize resource utilization. Operational scheduling for EV charging stations using dynamic charging-discharging rates is analyzed in [108]. By assessing real-time and forecast-based hosting capacity, grid operators can determine the best time slots for EV charging, battery storage charging/discharging, and demand response without

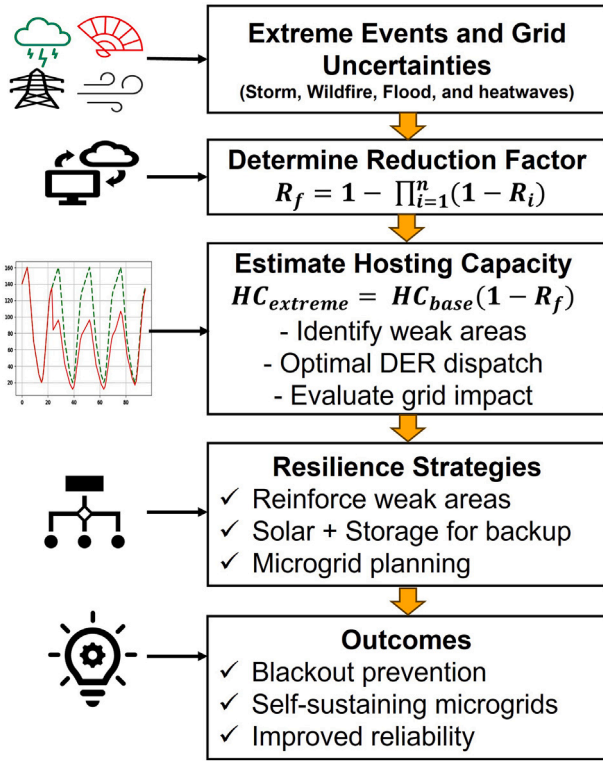


Fig. 11. Resilience improvement through hosting capacity estimation.

exceeding network constraints [109]. HC analysis provides the technical foundation to align market mechanisms with real-time network constraints [110]. Dynamic pricing [6,111] and setting regulatory policies [10] for DER integration can be implemented using hosting capacity analysis. In both cases, forward-looking hosting capacity estimation can inform the market operators and participants about possible tariff rates and upcoming changes. In addition, hosting capacity analysis aids local and wholesale energy markets by evaluating the feasibility of DER participation and flexible trading services [112].

3.5. Transmission-distribution coordination

This subsection explains how DER hosting capacity can be applied to the transmission-distribution coordination model. The current coordination model for transmission service operators (TSOs) and distribution system operators (DSOs) relies on three models: TSO-managed, DSO-managed, and hybrid [113]. TSO-managed coordination lacks DSO awareness, downstream visibility, and requires high data transfer, while the DSO-managed model faces market fragmentation, transmission-level risks, and limited system-wide visibility. To comprehend a win-win situation, hybrid coordination models are gaining popularity that employ market, operational, and grid management strategies [114,115].

Several studies have focused on enhancing flexibility in TSO-DSO coordination, addressing operational and market challenges. [116] proposes a DER and energy storage coordination scheme that minimises over-parameterisation and dynamically adjusts to TSO preferences, while Jiang et al. [117] introduce a market-based flexibility clearing strategy for energy transactions. Wang et al. [118] evaluate EV flexibility for optimizing ancillary services. These strategies highlight the need for hosting capacity analysis to support the integration of flexibility in coordination. However, challenges related to data granularity and latency in hybrid coordination can be addressed using smart metering, edge computing, and federated learning [113]. All these

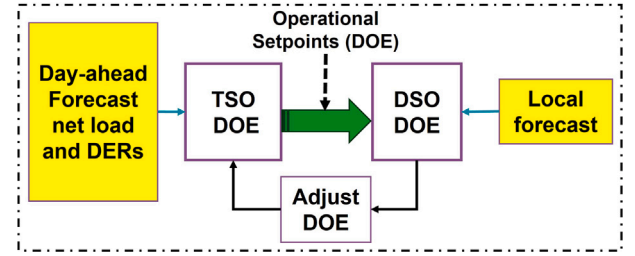


Fig. 12. Proposed hybrid TSO-DSO coordination using forecast-based DOEs.

methods require quantitative visibility into downstream distribution networks, which can serve as a foundational metric for optimizing TSO-DSO coordination. Hosting capacity or dynamic operating envelope can provide downstream visibility to the coordination models. These coordination models require specific features: selecting an appropriate scheme, estimating the flexibility region, and sufficient communication facilities.

Currently, the Australian Energy Market Operator (AEMO) uses forecasted load and generation profiles for their dispatch. While distribution networks increasingly use DOEs to manage network integrity, these limits are rarely shared with AEMO in aggregated form. As a result, AEMO continues to rely on top-down forecasting and static distribution feeder models to schedule generation and maintain system security [119]. Lacking downstream detailed visibility, their forecasts often fail to account for load and generation, leading to unintended events such as minimum demand and maximum export [120]. This gap in downstream visibility forces AEMO to infer the behavior of millions of DERs, leading to unpredictable events, security risks, and less efficient market dispatch. As DOE analysis practices and popularity grow, the use of DOEs in these coordination models will streamline the process and reduce duplication in flexibility calculations for operational or coordination models.

Fig. 12 shows a proposed TSO-DSO coordination process considering DOEs in both transmission and distribution systems. The proposed TSO-DSO coordination framework has major workflows, which are as follows-

- 1) TSO collects day-ahead forecasts of net load and aggregates DER output at each transmission connection point.
- 2) Based on these forecasts and transmission security constraints, the TSO computes TSO DOEs, i.e., allowable active/reactive power ranges at each T-D boundary bus.
- 3) Local forecasting of demand and DERs at the DSO site and development of distribution-level DOEs.
- 4) Sharing the distribution-level DOEs, aggregating DOEs into the upstream transmission-level, and publishing/adjusting the feasible operating region through operating setpoints.
- 5) The adjusted DOEs will be shared with the stakeholders involved in the energy markets.

This approach ensures the active participation and transparency between transmission and distribution operators by sharing consistent DOE information upstream and downstream. It improves DER utilization by applying coordinated, forward-looking limits rather than relying on emergency curtailment. This coordination increases the reliability of market dispatch, strengthens stakeholders' confidence by revealing available export headroom, and lays the groundwork for future decentralized flexibility markets. However, several practical challenges remain. Data sharing between transmission and distribution operators is still limited, and there is no unified or standardized DOE calculation methodology across operators. DER behavior is highly variable, and the absence of detailed network models and smart-meter visibility constrains accurate envelope calculation [73,121]. Additionally,

operational coordination and market dispatch using DOEs are still evolving.

4. Future research directions

Although there has been substantial growth in hosting capacity analysis in the literature and industry projects, several challenges remain in DER-rich power systems. Sections 2.4 and 2.5 discuss challenges and underlying causes, which are relevant for future research directions. An important consideration for standardizing hosting capacity definition is still missing in the literature [26,34]. The improvement in methods and tools, addressing accuracy and complexity issues, is still ongoing. As the technological and operational methods often involve costly adjustments and technology inclusions, prioritizing planning-level solutions for hosting capacity improvements is vital to reduce future retrofits and protect stakeholder benefits [5]. This section presents only the specific, application-oriented problems that should be the focus of future research and are discussed.

- **Integrated Transmission–Distribution Hosting Capacity and Coordination:** As the hosting capacity concept increased significantly on the distribution side, a unified T-D hosting capacity framework is important that jointly considers feeder-level constraints, bulk system stability, and interface limits. Current HC analysis practices treat transmission and distribution separately due to differences in voltage levels and visibility. However, high DER penetration requires coordinated modelling of net-demand variability, interface power exchanges, and DER flexibility. This integrated assessment can reveal hidden hosting potential and prevent conflicting operational decisions across network layers. Additionally, this coordination framework can be used for potential market dispatch. The key research questions that may be considered in future research are as follows-

- 1) How can distribution-level hosting capacity and granularity be used in transmission security-constrained hosting capacity assessment?
- 2) What is the minimum visibility granularity required for TSOs to incorporate downstream hosting limits into operational decision-making?
- 3) How can a feasible operating envelope be allocated at the transmission level for downstream distribution network operators?

- **HC-based Remuneration and Market Participation Models:** Most HC assessments remain technical and do not address the stakeholder economic perspectives, especially for prosumers and aggregators. Future research should develop market participation frameworks involving HC constraints-based locational pricing, flexibility procurement, and market trends. This includes developing remuneration mechanisms for network support services, such as voltage support, congestion management, fast frequency response, and compliance with export curtailment. HC-aware markets could also allocate costs more fairly and reward prosumers who contribute to increased system hosting capacity. The research questions that can be handled in future research are as follows-

- 1) How can prosumers be remunerated for providing network support services based on local HC constraints fairly and equitably?
- 2) How can HC metrics be incorporated into market bids/dispatch/locational pricing mechanisms to reflect network stress?

- **Real-time HC Estimation/Dynamic Hosting Capacity:** Dynamic HC requires probabilistic envelopes that adapt to forecasts, uncertainty and DER variability. Current HC methods in distribution address these issues; however, in transmission, current methods are static or scenario-based, which cannot capture rapid DER fluctuations and operational uncertainty. Future studies should focus on

dynamic hosting capacity models that can update export/import limits in real time or near-real time using measurements, forecasts, state estimation, and adaptive control strategies. Research questions may include-

- 1) What forecasting accuracy or resolution is required for dynamic hosting capacity to remain reliable under rapid DER fluctuations?
- 2) How can dynamic hosting capacity be synchronized across transmission and distribution, so that changes at one level do not offer new constraints at the other?
- 3) How will the probabilistic scenarios be developed in transmission-level dynamic hosting capacity estimation?
- 4) Which control strategies may provide the greatest marginal improvement in dynamic hosting capacity for transmission and distribution levels?

- **Cross-DER Interactions in Distribution Networks and Bulk Generation at Transmission Nodes:** Future HC research should examine the interactive and compound effects of multi-DER environments, in which rooftop PV, EV charging, battery storage systems, and industrial customers (either generation or load) collectively shape DER hosting limits at the T-D interface. A particular problem is that the Duck Curve is frequent in Australia's network, increasing network challenges and reducing hosting headroom for DERs at T-D networks. Therefore, understanding how combined DER behaviours modify the depth and timing of minimum demand, and how coordinated DER operation can smooth these effects, will be crucial for designing HC frameworks aiming for high-DER and low-demand conditions.

- **Hosting Capacity Under Network Resilience and Extreme Events:** Usually, hosting capacity estimation and enhancement are targeted for normal operating conditions, which should incorporate resilience considerations associated with heatwaves, bushfires, storms, floods, and other natural hazards. Future work must consider HC analysis under stressed conditions, evaluate the role of DERs in resilience support, and explore HC-based metrics for islanded operation, recovery, and black-start capabilities. This also requires stochastic and stress-withstanding frameworks to determine DER hosting headroom while maintaining robustness under disruptive events.

5. Conclusion

This paper reviews state-of-the-art hosting capacity analysis studies from the last decade, focusing on transmission-distribution networks to aid researchers, network operators, and planners in future augmentation. Although previous reviews have emphasized methods, tools, and technologies, the practical applications of hosting capacity outcomes for network operators and planners have been absent. The several takeaways from this review are as follows:

- This review identifies challenges in hosting capacity analysis, including complexity in network modeling, variability of DER and loads, selecting critical constraints and scenarios, and lack of standardized methods.
- Existing methods are broadly categorized into deterministic, stochastic, optimization-based, and streamlined approaches, each with trade-offs between accuracy, data needs, and computational complexity.
- Appropriate forecasting horizon and region-specific evaluation of DER and load profiles may improve accuracy and avoid under- or overestimation of hosting capacity.
- This review identifies a lack of transmission-level hosting capacity studies addressing multi-DER and transient impacts, signaling the need for transmission-distribution coordination. Despite their complexity with time-domain simulations and scenario selection, customizable tools can be utilized to achieve the best outcomes.

- Hosting capacity analysis requires a unified transmission-distribution perspective, as emerging DER-driven challenges span across network layers. Coordinated constraints, shared visibility, and T-D aligned operational frameworks are essential for accurate HC estimation, reliable dispatch, and future decentralized flexibility markets.
- Hosting capacity serves as a cross-functional tool with practical applications in planning upgrades, real-time operations, prosumer management, extreme event resilience, and transmission-distribution coordination.

Future research directions for advancing hosting capacity studies that extend to transmission networks, coupled with downstream customer-sited feeders, are also discussed in Section 4. By highlighting best practices and emerging trends, this review supports utilities, policymakers, and researchers in decision-making and optimizes grid planning and operations with large-scale DER integration.

CRediT authorship contribution statement

Ashish Kumar Karmaker: Writing – original draft, Visualization, Resources, Methodology, Formal analysis, Conceptualization. **Yang Du:** Writing – review & editing, Supervision, Conceptualization. **Jiajia Yang:** Writing – review & editing, Supervision. **Mohan Jacob:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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